

Combined Power-Water Flow Analysis of Regional Integrated Electricity and Water Networks

Xinyi Li

State Key Laboratory of Power
Transmission Equipment & System
Security and New Technology
Chongqing University
Chongqing, China
li3xy@mail.uc.edu

Xia Zhao

State Key Laboratory of Power
Transmission Equipment & System
Security and New Technology
Chongqing University
Chongqing, China
zx@cqu.edu.cn

Lun Yang

State Key Laboratory of Power
Transmission Equipment & System
Security and New Technology
Chongqing University
Chongqing, China
yanglun@cqu.edu.cn

Abstract—Multi-energy system has been a rising concept in recent years and there has been considerable research work on combined analysis, planning and operation of integrated power, natural gas and heating networks to explore the potential synergic benefits from multi-energy utilization. However, the integration of electricity and water networks is seldom considered in existing researches although these two networks are intrinsically coupled by components such as electric-motor-driven pumps (E-pumps). This paper, therefore, proposes to study the combined power-water flow analysis of integrated electricity and water networks (IEWN). With a focus on the regional IEWN composed of unbalanced power distribution system and municipal water network, the power-water flow problem is formulated with E-pumps being the coupling component and then the problem is solved utilizing Newton-Raphson method and Hardy-Cross method. Numerical results from a testing system that composes of a 33-bus distribution system and a 9-node water system are used to verify the integrated power-water flow analysis proposed.

Keywords— multi-energy system, integrated electricity and water network, power-water flow analysis, power flow, water flow

I. INTRODUCTION

In recent years, multi-energy system or integrated energy system (IES) has been a rising concept and drawn increased research interest [1]. There have been considerable research work on combined analysis, planning and operation of integrated power, natural gas and heating networks [2]-[4]. However, the integration of electricity and water networks is seldom considered in existing researches.

Generally, electricity networks and water networks are designed and operated independently as separated systems. In fact, these two networks are intrinsically interdependent due to the widely used electric-motor-driven pumps (E-pumps) in water networks, the significant effects of cooling water systems on the operation of thermal or nuclear power plants, and so on. Water pumps are important devices in water networks to raise water to a higher ground or floor for water supply and most of them are E-pumps. Since these pumps convert electrical energy to mechanical energy to provide pressure gain, they act as a coupling component between electricity and water networks. Moreover, according to the statistics in [5], in the United State, water facilities for drinking and waste water account for approximately 4% of overall energy consumption, of which 95%-98% is consumed by E-pumps, while in China, E-pumps, which involve in water supply, irrigation, cooling systems, etc., represent at least 21% of the country's total energy consumption [6]. Therefore, optimizing the operation of

electric-motor-driven pumps will have significant effects on both electricity and water networks, which shines a light on the possibility of synergistic operation of both networks and thus motivates us to study electricity and water networks in an integrated way.

Similar to studies on electricity networks, a lot of research efforts have been devoted to water network modeling, analysis, planning and operation. For example, He *et al.* proposed an improved iterative method for head-driven analysis to simulate abnormalities in water networks [7]. Ghelichi *et al.* proposed a novel robust optimization approach considering the uncertainty of water demand and rainfall [8]. However, the integration of the two networks is seldom discussed as listed above and only a few studies have focused on this aspect. Oikonomou *et al.* proposed a model to optimize the energy flexibility of water distribution system in power system [5]. Zamzam *et al.* studied the optimal power-water problem to minimize the operational cost of water system during a period considering both power flow constraints and hydraulic head constraints [9]. Therefore, integrated studies of the electricity-water network, especially the interaction between the two networks, are areas that needs further exploration.

In this paper, only electric-motor-driven water pump will be considered as a coupling component between electricity and water network, and electric-motor-driven water pump will be simply referred to as pump or water pump below.

This paper proposes to study the combined power-water flow problem of integrated electricity and water networks (IEWN). With a focus on regional IEWNs composed of power distribution system and municipal water networks, the combined power-water flow problem is formulated considering unbalanced three-phase distribution systems and E-pump as the coupling component. The power-water flow problem is then solved by two approaches, i.e., the approach using Newton-Raphson method to deal with both power and water flow and the other approach that uses Newton-Raphson to solve the three-phase power flow problem and Hardy-Cross method to solve the water flow problem. Two testing networks, i.e., a 9-node water network and a 15-node water network, are used to evaluate the performance of Newton-Raphson method and Hardy-Cross method for water flow problems. An integrated network composed of the 9-node water network and a 33-bus distribution system, is used to verify the integrated power-water flow analysis proposed.

The rest of this paper will be presented as follows. Section II presents the model of the regional IEWN and formulates the combined power-water flow problem. Section

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III describes the basic iteration of Newton-Raphson method and Hardy-Cross method for power and water flow analysis and then the overview of solving the power-water flow problem. Numerical results from the two testing networks are presented and analyzed in Section IV and conclusions are drawn in Section V.

II. INTEGRATED ELECTRICITY AND WATER NETWORK

A. Power Distribution System

Unlike electrical transmission system, power distribution system is unbalanced due to the existence of unbalanced loads and unbalanced distribution line parameters. In addition, growing distributed generation and a great number of single-phase loads, such as electrical vehicles, single-phase-rectifiers and electric-arc furnaces, are aggravating the situation. Failing to considering the unbalanced characteristics will result in great computational errors. Therefore, a three-phase power distribution system model is adopted in this paper.

Neglecting shunt capacitor, the voltage drop along a transmission line in a three-phase power distribution system is calculated by [10]:

$$\begin{bmatrix} \dot{U}_i^a \\ \dot{U}_i^b \\ \dot{U}_i^c \end{bmatrix} - \begin{bmatrix} \dot{U}_j^a \\ \dot{U}_j^b \\ \dot{U}_j^c \end{bmatrix} = \begin{bmatrix} z_{ij}^{aa} & z_{ij}^{ab} & z_{ij}^{ac} \\ z_{ij}^{ba} & z_{ij}^{bb} & z_{ij}^{bc} \\ z_{ij}^{ca} & z_{ij}^{cb} & z_{ij}^{cc} \end{bmatrix} \begin{bmatrix} \dot{I}_{ij}^a \\ \dot{I}_{ij}^b \\ \dot{I}_{ij}^c \end{bmatrix} \quad (1)$$

where $\dot{U}_i^\varphi$ and $\dot{U}_j^\varphi$ are the voltage at the two endpoints of line (i, j) , $\dot{I}_{ij}^\varphi$ is the three-phase current along the line ($\varphi \in B$, $B = \{a, b, c\}$), $z_{ij}^{\varphi\varphi}$ and $z_{ij}^{\varphi\psi}$ are self and mutual impedance of the line ($\psi \in B$, $\varphi \neq \psi$).

Let P_i^φ and Q_i^φ denote the active and reactive power at node i , the power balance equations for n -bus three-phase systems are [10]:

$$\begin{cases} P_i^\varphi = |\dot{U}_i^\varphi| \sum_{j=1}^n \sum_{\varphi=a}^c |\dot{U}_j^\varphi| (G_{ij}^{\varphi\varphi} \cos \theta_{ij}^{\varphi\varphi} + B_{ij}^{\varphi\varphi} \sin \theta_{ij}^{\varphi\varphi}) \\ Q_i^\varphi = |\dot{U}_i^\varphi| \sum_{j=1}^n \sum_{\varphi=a}^c |\dot{U}_j^\varphi| (G_{ij}^{\varphi\varphi} \sin \theta_{ij}^{\varphi\varphi} - B_{ij}^{\varphi\varphi} \cos \theta_{ij}^{\varphi\varphi}) \end{cases} \quad (2)$$

B. Water Distribution Network

The topology of a water distribution network (WDN) is a network composed of nodes and links. In terms of components, a water distribution network consists of reservoirs, tanks, pumps, valves, and pipes.

- **Reservoirs:** A reservoir is the source of water that is assumed to be able to provide infinite water or accept excessive water in water flow analysis. A reservoir may be a lake or a groundwater well.
- **Tanks:** A tank serves as a storage facility when there is a low water demand, or a finite source of water when water is in high demand. In steady-state models, a tank would only play the role of either storage or water source at one time, but in extended-period simulation models, it may change its role because an overflow or exhaust is not allowed in reality.

- **Pipe:** Pipe is the linkage between nodes and pipeline is assembled in series using multiple pipes. In steady-state models, the characteristics of pipes are assumed to be the same along the pipeline and minor losses caused by configuration and appurtenance are ignored in this paper. The length, diameter and roughness coefficient of pipes are needed to determine the resistance of pipes and then the pressure loss.

The two state variables of a water network are hydraulic head H and flow rate q . Flow rate q is also known as volumetric flow rate in hydrology, which represents the amount of water that flows through a pipe per unit time. The SI unit of flow rate is m/s. Head is the energy of water per unit weight, which is due to its velocity, elevation and pressure. In a practical water network, the velocity head is usually much less than elevation head and pressure head so can be neglected. Therefore, the hydraulic head H can be determined as [11]:

$$H = \frac{P}{\rho g} + z \quad (3)$$

where P is the pressure, ρ is the density of water, g is the gravity acceleration and z is the elevation above a preset datum. In water flow analysis, there must be a reference node, of which the hydraulic head is known and sometimes constant, so that the head at other nodes can be computed. Mostly, the reference node is a reservoir or a tank when there is no reservoirs.

Due to pipe friction, there will be head loss through a pipe. The direction of head drop agrees with the direction of flow. Basically, head loss h_{ij} is proportional to the flow rate q_{ij} to the n^{th} power along the pipe (i, j) [11]:

$$h_{ij} = r q_{ij}^n \quad (4)$$

where r is the resistance coefficient of pipe, which can be calculated by the Hazen-Williams formula with $n=1.852$ [11]:

$$r = \frac{10.67l}{C^{1.852} d^{4.87}} \quad (5)$$

where l and d denote the length and diameter of the pipe respectively, C is the pipe roughness coefficient, determined by the pipe's material and condition.

In a water network without any loops, the water flows in each pipe and heads at each node are constant, while that of a complex network, of which the topology is not simply radiant, are unknown to us. The purpose of water flow analysis is to determine the distribution of water flows in a water network so that total head loss of the network can be obtained and that further system operations can be performed. The water flow analysis is based on mass and energy conservation:

- **Mass conservation:** For each junction in a water network, the water mass that flows into the junction must be equal to the water mass that flows out of the junction at any time. Let Q_j denote the water demand at junction j , q_{ij} and q_{jk} denote the water flow entering and leaving junction j respectively, junction i and k represent the nodes connected to j , the mass conservation law is expressed as [11]:

$$\sum q_{ij} - \sum q_{jk} = Q_j \quad (6)$$

- Energy conservation: For any two junctions connected by a pipeline, the head difference is the same no matter what path is taken. In other words, with head losses in clockwise direction being positive and head losses in counter-clockwise direction being negative, the sum of head losses around a loop is 0 [11]:

$$\sum h_{ij} = 0 \quad (7)$$

where h_{ij} is the head loss of pipe (i, j) that is a part of a loop.

C. Coupling Between Power System and Water Network

Pumps and pumping stations are the connection points between power system and water network, which consume electricity to provide pressure gain for water distribution network to overcome elevation difference or to meet the head demand of consumers. Generally, there are more than one pump in a pumping station in case of inadequate water flows and fire emergency. Various configurations of pumps would enhance the flexibility of water supply and would also increase the efficiency of pumps.

Suppose a pump p is installed on pipe (i, j) , the pressure gain $h_{p,ij}$ (also called pump head) is determined by pump's discharge $q_{p,ij}$ [11]:

$$h_{p,ij} = Aq_{p,ij}^2 + Bq_{p,ij} + C \quad (8)$$

where A , B and C are coefficients determined by pump characteristics. This head curve is one of the pump characteristic curves and the actual head curve of a pump needs to be developed by pump performance test. If two pumps are running in parallel, the head curve of this configuration can be obtained by combining the original two curves. That is, with the same pressure gain, the output discharge is the sum of the two pumps' flow.

In addition to flow rate, motor speed would also affect the pressure gain of a pump. Let s denote the ratio of actual motor speed to nominal motor speed, now (8) is [9]:

$$h_{p,ij} = Aq_{p,ij}^2 + Bq_{p,ij}s + Cs^2 \quad (9)$$

This leads to the utilization of variable-speed pumps. Compared with fixed-speed pumps, variable-speed pumps have multiple head curves corresponding to different motor speeds. Variable-speed pumps are more flexible and applicable in reality since variable-speed pumps can adjust motor speed to output desired pressure or flow, which is a significant way to increase pump efficiency.

Most electric pumps are driven by three-phase motors except the smallest ones. As a load in power system, the power consumed by a pump can be obtained by the following expression [9]:

$$P_{p,ij} = \frac{\rho g}{\eta_{p,ij}} h_{p,ij} q_{p,ij} \quad (10)$$

where η_p is the wire-to-water efficiency of pump, which is also the overall efficiency. Overall efficiency is the product of motor efficiency and pump efficiency. Motor efficiency is comparatively constant when the input power reaches some value while pump efficiency varies dramatically with flow rate. Theoretically, the overall efficiency is a quadratic

curve with the quadratic term being negative, but actual efficiency curve would be different from the theoretical one and also needs to be attained by pump performance test.

III. SOLVING METHODS FOR STEADY FLOW ANALYSIS

A. Newton-Raphson Method for Power Flow Analysis

Newton-Raphson method is one of the most popular iterative methods for steady-state power flow analysis. Its favorable characteristics (power flow results can be obtained in a few iterations given a proper initial guess) make Newton's method a practical and efficient method.

In order to solve the power equations, (2) is expended in Taylor series. Ignoring higher-order terms results in linear equations that can be solved easily. Let ΔP_i and ΔQ_i represent the active and reactive power at node i , the revising voltage magnitude $\Delta |\dot{U}_i|$ and voltage angle $\Delta \delta_i$ at node i are obtained by [12]:

$$\begin{bmatrix} \Delta P_i \\ \Delta Q_i \end{bmatrix} = J \begin{bmatrix} \Delta \delta_i \\ \Delta |\dot{U}_i| \end{bmatrix} \quad (11)$$

where J represents Jacobian matrix:

$$J = \begin{bmatrix} \frac{\partial \Delta P_i}{\partial \delta_i} & \frac{\partial \Delta P_i}{\partial |\dot{U}_i|} \\ \frac{\partial \Delta Q_i}{\partial \delta_i} & \frac{\partial \Delta Q_i}{\partial |\dot{U}_i|} \end{bmatrix} \quad (12)$$

B. Newton-Raphson Method for Water Flow Analysis

Newton-Raphson method is also applicable to water flow analysis. Applying Newton's method, if there are L loops in a water distribution network, the water flow solution is:

$$[\Delta h_m] = -J[\Delta q_m] \quad (13)$$

$$J = \begin{bmatrix} \frac{\partial \Delta h_m}{\partial \Delta q_m} \end{bmatrix} \quad (14)$$

where Δh_m is the total head loss around loop m ($m=1, 2, \dots, L$), and Δq_m is the corresponding counterbalancing flow.

C. Hardy-Cross Method for Water Flow Analysis

In 1936, Hardy Cross developed a method to determine the water flow distribution of water networks. Hardy-Cross method is an iterative method which balances heads at junctions by revising the flows through pipes in each loop. Hardy's method neglects the influence of adjacent loops on revised flow when balancing heads for each loop. Assume there are L loops in a water network, the revising flow rate for loop m is [13]:

$$\Delta q_m = -\frac{\Delta h_m}{n \sum |rq_i^{n-1}|} \quad (15)$$

where Δq_m is the counterbalancing flow for all pipes that form loop m ($m=1, 2, \dots, L$), and Δh_m is the total head loss around the loop m , q_i represents the flow rate through the pipes in loop m .

Hardy-Cross method can be regarded as a simplified version of Newton's method, since they are both iterative

methods that use Taylor expansion for linearization, but Hardy's method uses only the diagonal elements of Jacobian matrix. Thus, there is no need for matrix computation when obtaining counterbalancing flows, which makes Hardy's method favorable for manual calculation. Besides, this simplification may lead to a slower convergence that will be discussed in case study. The procedure for solving water flow of a pipeline network using Hardy Cross method is summarized as follows:

- 1) Initialize all the water flows of pipeline network, which satisfies the mass conservation at each junction.
- 2) Compute the head losses of each pipe and then the total losses around every loop.
- 3) Compute the counterbalancing flows of each loop and then the revised flows of each pipe.
- 4) Update water flow rates and head losses along all pipes. With flows in clockwise direction being positive and flows in counter-clockwise direction being negative, the updated flow rate of a pipe equals the sum of counterbalancing flow rate and original flow rate.
- 5) Repeat the 2nd and 3rd step if the total head losses of each loop are greater than the preset tolerance; otherwise, stop the iteration then water flows are obtained.

D. Solving Integrated Electricity-Water Network Model

The power-water flow of IEWN is obtained by solving power and water flow equations, i.e., (2), (6) and (7). Fig. 1 demonstrates the process of solving the proposed IEWN model.

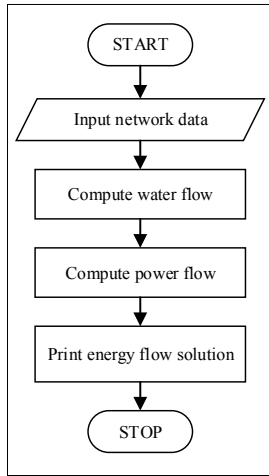


Fig. 1. Procedures of solving power and water flow.

First, information of both power and water system is collected, including power loads, water loads, pump curves, pipeline parameters and transmission line parameters. Since we are dealing with a steady-state model and pumps serve as loads in power system, water flow is obtained using Hardy-Cross method or Newton-Raphson method before anything else. If water flow solution converges, pumps are converted to electrical loads in power system; if not, perform water flow analysis with more proper initial flow rates. Then power flow can be obtained applying Newton's method.

IV. CASE STUDIES

The proposed IEWN model is verified in this section. There are two cases: the first one compares the performance

of Newton-Raphson method and Hardy-Cross method for steady-state water flow analysis on two different WDNs and the second testifies the proposed model on an IEWN that consists of an unbalanced 33-bus system and a 9-node water distribution network. Simulations are run on a laptop computer with an Intel i7-7700 CPU and 8GB RAM.

A. Case I: Evaluation of Hardy-Cross and Newton-Raphson Method for Water Flow Analysis

Two different WDNs are used to evaluate the performance of Hardy-Cross method and Newton-Raphson method, as shown in Fig. 2 and Fig. 3. WDN1 consists of 9 nodes and 2 loops, one is a closed loop and the other is a pseudo loop. WDN2 consists of 15 nodes, 6 closed loops and 2 pseudo loops. The flow distribution of each network is computed applying Hardy's method and Newton's method separately with tolerance set to 10^{-6} . The flow rates and heads obtained by Newton's and Hardy's method are identical. The number of iteration and computational time are shown in Table I and II.

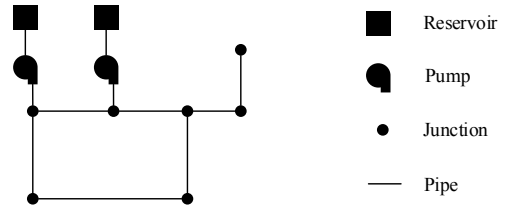


Fig. 2. Network topology of WDN1.

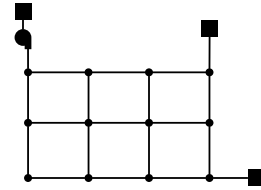


Fig. 3. Network topology of WDN2.

TABLE I. NUMBER OF ITERATIONS USING THE TWO METHODS

Method	Number of Iterations	
	WDN1	WDN2
Hardy-Cross	9	72
Newton-Raphson	4	7

TABLE II. COMPUTATIONAL TIME USING THE TWO METHODS

Method	Computational Time(s)	
	WDN1	WDN2
Hardy-Cross	6.2651e-04	0.0098
Newton-Raphson	7.6172e-04	0.0016

As seen from Table I, the solution of Hardy-Cross method converges more slowly than that of Newton's method, which is because when computing revised flow for a loop, Hardy's method neglects the influence of its adjacent loops. From WDN1 to WDN2, the number of iterations that Hardy-Cross method needs for convergence increases notably while the number that Newton's method needs increases slightly. Moreover, the computational time in Table II for each WDN utilizing the two methods. When obtaining the flow distribution in WDN1, a simple network with only two loops, it takes Newton's method more time than Hardy-

Cross method; however, it is the opposite when we compute the flow distribution of WDN2, which is a relatively complex network. The increment of computational time for Hardy-Cross method is remarkably larger than that of Newton's method. During multiple tests, it is also found that when Hardy-Cross method is used, the convergence of solution is more sensitive to pipe flow rates assigned initially while Newton's method is less sensitive.

B. Case II: Combined Power-Water Flow Analysis of Integrated Electricity-Water Network

The testing integrated electricity-water network consists of an unbalanced 33-bus system [14] and a modified 9-node WDN from [14], as shown in Fig. 4. There are two reservoirs and three electric pumps in the WDN, where pump 1 is a variable-speed pump that has a constant discharge. Water is delivered through 9 pipes to consumers at node 3-7.

Assume all pumps are driven by three-phase electric motors, of which the motor efficiency is kept at 80%. Combined power-water flow is attained using Newton's method and Hardy's method. Numerical results, including voltage magnitude of each bus in power system, and hydraulic head of nodes in WDN, are presented in Fig. 5 and Table III.

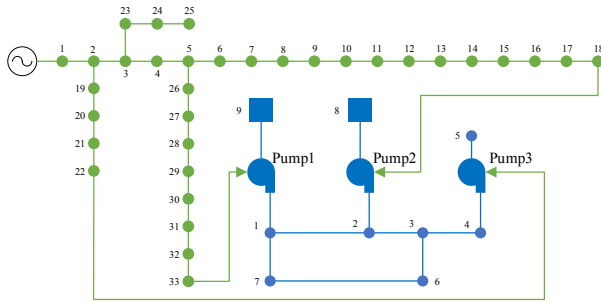


Fig. 4. Schematic diagram of the integrated electricity-water network.

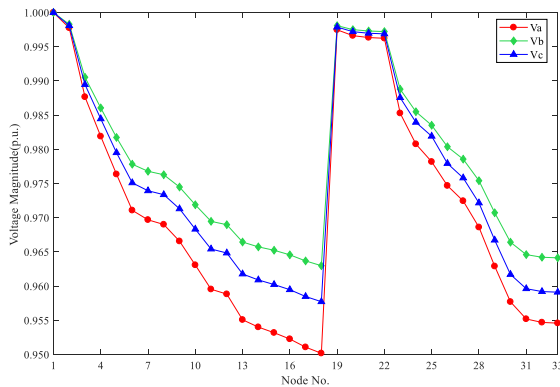


Fig. 5. Voltage magnitude of the 33-bus power distribution system.

TABLE III. HEAD RESULTS OF WATER DISTRIBUTION NETWORK

Bus No.	Head of Nodes		
	Elevation(m)	Pressure(m)	Hydraulic head(m)
1	5.0	53.6129	58.6129
2	5.0	54.1817	59.1817
3	0	50.9232	50.9232
4	0	45.3544	45.3544
5	35.0	41.6739	76.6739

Bus No.	Head of Nodes		
	Elevation(m)	Pressure(m)	Hydraulic head(m)
6	7.5	39.8934	47.3934
7	12.0	41.5432	53.5432

V. CONCLUSION

This paper proposes to study the integrated electricity-water network and conducts combined power-water flow analysis using Newton-Raphson method and Hardy-Cross method. Newton's method and Hardy's method are both applicable to water flow analysis and applying the two methods generates identical water flow results. The number of iteration and computational time reveal that Newton's method has a better performance than Hardy's method in steady-state water flow analysis, especially when dealing with complex pipeline networks. In addition, steady-state combined power-water flow for power distribution system and municipal water network is obtained from the testing system, which would be a start of exploration into integrated electricity-water network.

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